

# GCP Datastream gcloud Operations Guide

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**Status:** Reference

## Original User Input

Documentation of how to delete, pause, restart and create a Datastream instance using gcloud CLI.

Sources: AlloyDB and Oracle on-prem. Destination: Cloud Storage.

## Summary

Complete reference for managing GCP Datastream streams and connection profiles via `gcloud` CLI. Covers authentication, reading existing configurations, and the full lifecycle for two scenarios:

- **AlloyDB** → **Cloud Storage** (via VPC peering private connection)
- **Oracle on-prem** → **Cloud Storage** (via PSC private connection)

All examples use real configurations from project `prj-cal-w-wl5-t-6c00-53ad` in `europa-west3`.

## 1. Authentication

Before running any `gcloud datastream` command, you must be authenticated and have the correct project set.

```
# Authenticate (opens browser)
gcloud auth login

# Verify active account
gcloud auth list

# Set the target project
gcloud config set project prj-cal-w-wl5-t-6c00-53ad

# Verify
gcloud config get-value project
```

If your token expires mid-session, gcloud returns:

```
ERROR: Reauthentication failed. cannot prompt during non-interactive execution.
```

Re-run `gcloud auth login` to refresh.

## 2. Reading Existing Configurations

Before creating new streams or connection profiles, read the existing ones to understand the current setup and reuse them as templates for new instances.

### List What Exists

```
# List all streams
gcloud datastream streams list --location=europe-west3

# List all connection profiles
gcloud datastream connection-profiles list --location=europe-west3

# List private connections
gcloud datastream private-connections list --location=europe-west3
```

# Export Configurations as JSON Templates

```
# Export a stream config
gcloud datastream streams describe STREAM_ID \
  --location=europe-west3 --format=json > stream-template.json

# Export a connection profile config
gcloud datastream connection-profiles describe PROFILE_ID \
  --location=europe-west3 --format=json > profile-template.json
```

## Current Inventory (Test Environment)

### Streams:

| Stream ID                                | Source                    | State       |
|------------------------------------------|---------------------------|-------------|
| orauat-1060-bucket                       | Oracle<br>TMS1060.SENDUNG | RUNNING     |
| new-dispo-cdc-datastream-sendung-abn1034 | AlloyDB tms1034.sendung   | RUNNING     |
| new-dispo-cdc-datastream-sendung-abn2820 | AlloyDB tms2820.sendung   | NOT_STARTED |

### Connection Profiles:

| Profile ID                                       | Type       | Target                             |
|--------------------------------------------------|------------|------------------------------------|
| ora-datastream-1060uat                           | Oracle     | 10.32.0.71:1521<br>(TMSA)          |
| new-dispo-cdc-postgres-connection-abn1034-1      | PostgreSQL | 10.100.47.236:5432<br>(abn1034)    |
| new-dispo-cdc-postgres-connection-abn2820        | PostgreSQL | AlloyDB abn2820                    |
| new-dispo-cdc-cloud-storage-connection-abn1034-1 | GCS        | abn1043-sendung-<br>bucket-1       |
| new-dispo-cdc-cloud-storage-connection-uat2820   | GCS        | AlloyDB abn2820<br>bucket          |
| datastream-to-cloud-storage                      | GCS        | tms-alloydb-<br>datastream-bucket- |

| Profile ID | Type | Target  |
|------------|------|---------|
|            |      | wl5-t-t |

**Private Connections:**

| Connection ID                   | Type           | Details                                             |
|---------------------------------|----------------|-----------------------------------------------------|
| psc-datastream-t-wl5            | PSC            | Network attachment:<br>na-datastream-t-wl5 (Oracle) |
| datastream-connectivity-wl5-t-t | VPC<br>Peering | Subnet: 10.100.53.0/29 (AlloyDB)                    |

Use `describe` on any of these to get the full JSON config, then adapt it for a new instance.

### 3. Connection Profile Operations

#### 3a. Create AlloyDB Source Profile

AlloyDB uses `--type=postgresql` (PostgreSQL wire protocol). Connectivity uses VPC peering private connection.

```
gcloud datastream connection-profiles create NEW_PROFILE_ID \
  --location=europe-west3 \
  --type=postgresql \
  --display-name="DISPLAY_NAME" \
  --postgresql-hostname=ALLOYDB_IP \
  --postgresql-port=5432 \
  --postgresql-username=USERNAME \
  --postgresql-password=PASSWORD \
  --postgresql-database=DATABASE_NAME \
  --private-connection=datastream-connectivity-wl5-t-t
```

**Real example** (from `new-dispo-cdc-postgres-connection-abn1034-1`):

```
gcloud datastream connection-profiles create new-dispo-cdc-postgres-connection-abn1034-
--location=europe-west3 \
--type=postgresql \
--display-name="new-dispo-cdc-postgres-connection-abn1034-1" \
--postgresql-hostname=10.100.47.236 \
--postgresql-port=5432 \
--postgresql-username=tmsbr1034 \
--postgresql-password=PASSWORD \
--postgresql-database=abn1034 \
--private-connection=datastream-connectivity-wl5-t-t
```

| Flag                  | Description                                                |
|-----------------------|------------------------------------------------------------|
| --type=postgresql     | AlloyDB uses PostgreSQL protocol                           |
| --postgresql-hostname | AlloyDB instance IP (reachable via private connection)     |
| --postgresql-port     | Default 5432                                               |
| --postgresql-database | Database name (e.g. abn1034 )                              |
| --postgresql-username | Replication user (e.g. tmsbr1034 )                         |
| --postgresql-password | Or use --postgresql-secret-manager-stored-password         |
| --private-connection  | VPC peering connection ( datastream-connectivity-wl5-t-t ) |

### 3b. Create Oracle On-Prem Source Profile

Oracle uses --type=oracle . Connectivity uses PSC (Private Service Connect).

```
gcloud datastream connection-profiles create NEW_PROFILE_ID \
--location=europe-west3 \
--type=oracle \
--display-name="DISPLAY_NAME" \
--oracle-hostname=ORACLE_IP \
--oracle-port=1521 \
--oracle-username=USERNAME \
--oracle-password=PASSWORD \
--oracle-database-service=SERVICE_NAME \
--private-connection=psc-datastream-t-wl5
```

**Real example** (from ora-datastream-1060uat ):

```
gcloud datastream connection-profiles create ora-datastream-1060uat \
  --location=europe-west3 \
  --type=oracle \
  --display-name="ORA-datastream-1060UAT" \
  --oracle-hostname=10.32.0.71 \
  --oracle-port=1521 \
  --oracle-username=C##P3_LOGMINER \
  --oracle-password=PASSWORD \
  --oracle-database-service=TMSA \
  --private-connection=psc-datastream-t-wl5
```

| Flag                      | Description                             |
|---------------------------|-----------------------------------------|
| --type=oracle             | Oracle source                           |
| --oracle-hostname         | Oracle host IP (e.g. 10.32.0.71 )       |
| --oracle-port             | Default 1521                            |
| --oracle-database-service | Oracle service name (e.g. TMSA )        |
| --oracle-username         | Datastream user (e.g. C##P3_LOGMINER )  |
| --private-connection      | PSC connection ( psc-datastream-t-wl5 ) |

### 3c. Create Cloud Storage Destination Profile

```
gcloud datastream connection-profiles create NEW_PROFILE_ID \
  --location=europe-west3 \
  --type=google-cloud-storage \
  --display-name="DISPLAY_NAME" \
  --bucket=BUCKET_NAME \
  --root-path=
```

**Real example** (from new-dispo-cdc-cloud-storage-connection-abn1034-1 ):

```
gcloud datastream connection-profiles create new-dispo-cdc-cloud-storage-connection-abn
--location=europe-west3 \
--type=google-cloud-storage \
--display-name="new-dispo-cdc-cloud-storage-connection-abn1034-1" \
--bucket=abn1043-sendung-bucket-1 \
--root-path=/
```

| Flag                        | Description                      |
|-----------------------------|----------------------------------|
| --type=google-cloud-storage | GCS destination                  |
| --bucket                    | Target GCS bucket name           |
| --root-path                 | Path prefix (must start with / ) |

No credentials needed -- Datastream uses the project's default service account.

### 3d. List Connection Profiles

```
gcloud datastream connection-profiles list --location=europe-west3
```

### 3e. Delete a Connection Profile

```
gcloud datastream connection-profiles delete PROFILE_ID --location=europe-west3
```

A connection profile **cannot** be deleted while referenced by an active stream.

## 4. Stream Operations

### 4a. Create AlloyDB → Cloud Storage Stream

```
gcloud datastream streams create STREAM_ID \  
  --location=europe-west3 \  
  --display-name="DISPLAY_NAME" \  
  --source=ALLOYDB_PROFILE_ID \  
  --destination=GCS_PROFILE_ID \  
  --postgresql-source-config=alloydb-source-config.json \  
  --gcs-destination-config=gcs-dest-config.json \  
  --backfill=none
```

**alloydb-source-config.json** (based on `new-dispo-cdc-datastream-sendung-abn1034`):

```
{  
  "replicationSlot": "sendung_slot_abn1034",  
  "publication": "sendung_pub",  
  "includeObjects": {  
    "postgresqlSchemas": [  
      {  
        "schema": "tms1034",  
        "postgresqlTables": [  
          { "table": "sendung" }  
        ]  
      }  
    ]  
  }  
}
```

**gcs-dest-config.json** (based on existing streams):

```
{  
  "fileRotationInterval": "60s",  
  "fileRotationMb": 50,  
  "jsonFileFormat": {  
    "compression": "NO_COMPRESSION",  
    "schemaFileFormat": "NO_SCHEMA_FILE"  
  }  
}
```



### Backfill options:

- `--backfill=none` -- CDC only, no historical data
- `--backfill=all` -- backfill all existing rows, then switch to CDC

## 4b. Create Oracle → Cloud Storage Stream

```
gcloud datastream streams create STREAM_ID \  
  --location=europe-west3 \  
  --display-name="DISPLAY_NAME" \  
  --source=ORACLE_PROFILE_ID \  
  --destination=GCS_PROFILE_ID \  
  --oracle-source-config=oracle-source-config.json \  
  --gcs-destination-config=gcs-dest-config.json \  
  --backfill=all
```

**oracle-source-config.json** (based on `orauat-1060-bucket`):

```
{  
  "includeObjects": {  
    "oracleSchemas": [  
      {  
        "schema": "TMS1060",  
        "oracleTables": [  
          { "table": "SENDUNG" }  
        ]  
      }  
    ]  
  },  
  "maxConcurrentCdcTasks": 5,  
  "maxConcurrentBackfillTasks": 16,  
  "dropLargeObjects": {}  
}
```

**gcs-dest-config.json** (Oracle variant with GZIP compression):

```
{
  "path": "/UATDataStream",
  "fileRotationInterval": "60s",
  "fileRotationMb": 50,
  "jsonFileFormat": {
    "compression": "GZIP",
    "schemaFileFormat": "NO_SCHEMA_FILE"
  }
}
```

## 4c. GCS Destination Config Options

| Field                           | Description                                   |
|---------------------------------|-----------------------------------------------|
| path                            | Subdirectory under the bucket's root-path     |
| fileRotationMb                  | Rotate file after this size in MB (e.g. 50 )  |
| fileRotationInterval            | Rotate file after this duration (e.g. "60s" ) |
| jsonFileFormat.compression      | "NO_COMPRESSION" or "GZIP"                    |
| jsonFileFormat.schemaFileFormat | "NO_SCHEMA_FILE" or "AVRO_SCHEMA_FILE"        |
| avroFileFormat                  | Use {} for binary Avro instead of JSON        |

### GCS output path structure (automatic):

```
gs://BUCKET/ROOT_PATH/PATH/SCHEMA.TABLE/yyyy/mm/dd/hh/mm/filename
```

## 4d. Pause a Stream

```
gcloud datastream streams update STREAM_ID \
  --location=europe-west3 \
  --state=PAUSED \
  --update-mask=state
```

**State transition:** RUNNING -> DRAINING -> PAUSED

**Important:**

- Some in-flight data may still arrive at the destination after pausing
- If paused longer than the source database's WAL/redo log retention period, resuming may fail -- you would need to delete and recreate the stream

## 4e. Resume a Stream

```
gcloud datastream streams update STREAM_ID \  
  --location=europe-west3 \  
  --state=RUNNING \  
  --update-mask=state
```

**State transition:** PAUSED -> RUNNING

## 4f. Delete a Stream

```
gcloud datastream streams delete STREAM_ID --location=europe-west3
```

Add `--quiet` to skip confirmation prompt.

Deleting stops replication immediately. Cleanup on the source database is **not** automatic:

**AlloyDB / PostgreSQL:**

```
SELECT pg_drop_replication_slot('sendung_slot_abn1034');  
DROP PUBLICATION sendung_pub;
```

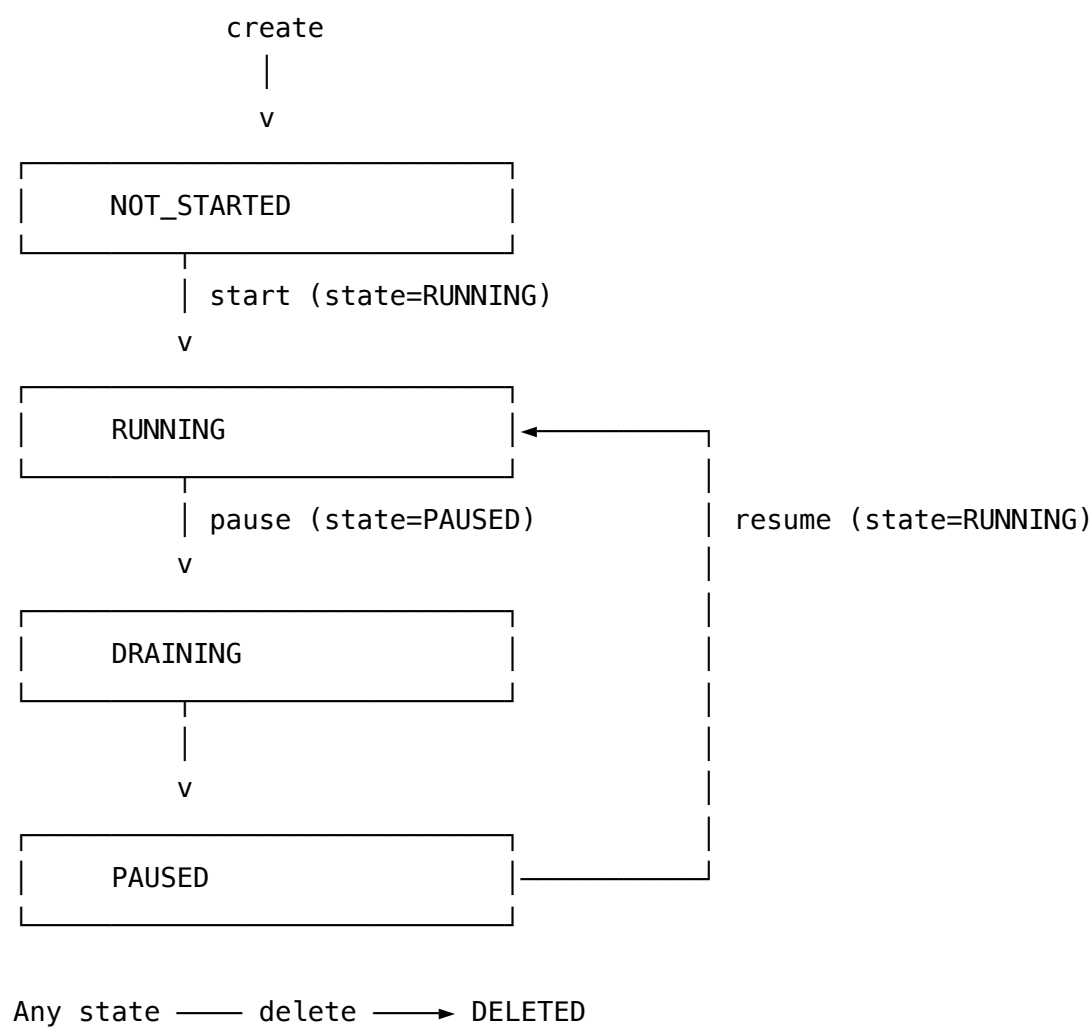
**Oracle:**

The LogMiner session is cleaned up by Datastream, but verify no orphaned redo log groups remain.

## 4g. Describe a Stream

```
gcloud datastream streams describe STREAM_ID \  
  --location=europe-west3 \  
  --format=json
```

# 5. Stream State Machine



# 6. Common Flags

All gcloud datastream commands support:

| Flag                 | Description                                   |
|----------------------|-----------------------------------------------|
| --project=PROJECT_ID | Target GCP project                            |
| --location=LOCATION  | Region (always europe-west3 for this project) |
| --format=FORMAT      | Output format: json , yaml , text             |
| --quiet              | Suppress confirmation prompts                 |

| Flag                           | Description                    |
|--------------------------------|--------------------------------|
| <code>--verbosity=LEVEL</code> | debug , info , warning , error |

## 7. Prerequisites

### AlloyDB Source

- Logical replication enabled: `wal_level = logical`
- A replication slot and publication must exist for the Datastream user
- User needs `REPLICATION` privilege
- Private connection via VPC peering must exist ( `datastream-connectivity-wl5-t-t` )

```
CREATE USER tmsbr1034 WITH REPLICATION LOGIN PASSWORD 'password';
GRANT USAGE ON SCHEMA tms1034 TO tmsbr1034;
GRANT SELECT ON ALL TABLES IN SCHEMA tms1034 TO tmsbr1034;
CREATE PUBLICATION sendung_pub FOR TABLE tms1034.sendung;
SELECT pg_create_logical_replication_slot('sendung_slot_abn1034', 'pgoutput');
```

### Oracle On-Prem Source

- Oracle 11g+ with LogMiner
- Supplemental logging enabled
- Datastream user needs LogMiner privileges (e.g. `C##P3_LOGMINER` )
- Private connection via PSC must exist ( `psc-datastream-t-wl5` )

### Cloud Storage Destination

- Target bucket must exist
- Datastream service account needs `Storage Object Admin` on the bucket

### General

- All resources (connection profiles, streams, private connections) must be in the **same location** ( `europa-west3` )

## 8. Quick Reference: Full Lifecycle

### AlloyDB → Cloud Storage

```
# 0. Authenticate
gcloud auth login
gcloud config set project prj-cal-w-wl5-t-6c00-53ad

# 1. Read existing configs as templates
gcloud datastream connection-profiles list --location=europe-west3
gcloud datastream streams describe new-dispo-cdc-datastream-sendung-abn1034 \
  --location=europe-west3 --format=json

# 2. Create connection profiles
gcloud datastream connection-profiles create new-dispo-cdc-postgres-connection-ABNXXXX \
  --location=europe-west3 --type=postgresql \
  --display-name="new-dispo-cdc-postgres-connection-ABNXXXX" \
  --postgresql-hostname=ALLOYDB_IP --postgresql-port=5432 \
  --postgresql-database=abnXXXX \
  --postgresql-username=tmsbrXXXX --postgresql-password=PASSWORD \
  --private-connection=datastream-connectivity-wl5-t-t

gcloud datastream connection-profiles create new-dispo-cdc-cloud-storage-connection-abnXXXX \
  --location=europe-west3 --type=google-cloud-storage \
  --display-name="new-dispo-cdc-cloud-storage-connection-abnXXXX" \
  --bucket=abnXXXX-sendung-bucket --root-path=/

# 3. Create stream
gcloud datastream streams create new-dispo-cdc-datastream-sendung-abnXXXX \
  --location=europe-west3 \
  --display-name="new-dispo-cdc-datastream-sendung-abnXXXX" \
  --source=new-dispo-cdc-postgres-connection-ABNXXXX \
  --destination=new-dispo-cdc-cloud-storage-connection-abnXXXX \
  --postgresql-source-config=alloydb-source-config.json \
  --gcs-destination-config=gcs-dest-config.json \
  --backfill=none

# 4. Pause
gcloud datastream streams update new-dispo-cdc-datastream-sendung-abnXXXX \
  --location=europe-west3 --state=PAUSED --update-mask=state

# 5. Resume
```

```
gcloud datastream streams update new-dispo-cdc-datastream-sendung-abnXXXX \  
  --location=europe-west3 --state=RUNNING --update-mask=state
```

# 6. Delete stream

```
gcloud datastream streams delete new-dispo-cdc-datastream-sendung-abnXXXX \  
  --location=europe-west3
```

# 7. Delete connection profiles

```
gcloud datastream connection-profiles delete new-dispo-cdc-postgres-connection-ABNXXXX \  
  --location=europe-west3
```

```
gcloud datastream connection-profiles delete new-dispo-cdc-cloud-storage-connection-abn \  
  --location=europe-west3
```

# Oracle On-Prem → Cloud Storage

# 0. Authenticate

```
gcloud auth login
gcloud config set project prj-cal-w-wl5-t-6c00-53ad
```

# 1. Read existing configs as templates

```
gcloud datastream connection-profiles describe ora-datastream-1060uat \
  --location=europe-west3 --format=json
gcloud datastream streams describe orauat-1060-bucket \
  --location=europe-west3 --format=json
```

# 2. Create connection profiles

```
gcloud datastream connection-profiles create ora-datastream-XXXX \
  --location=europe-west3 --type=oracle \
  --display-name="ORA-datastream-XXXX" \
  --oracle-hostname=ORACLE_IP --oracle-port=1521 \
  --oracle-username=C##P3_LOGMINER --oracle-password=PASSWORD \
  --oracle-database-service=TMSA \
  --private-connection=psc-datastream-t-wl5
```

```
gcloud datastream connection-profiles create gcs-oracle-XXXX \
  --location=europe-west3 --type=google-cloud-storage \
  --display-name="GCS Oracle XXXX" \
  --bucket=oracle-XXXX-bucket --root-path=
```

# 3. Create stream

```
gcloud datastream streams create oraXXXX-bucket \
  --location=europe-west3 \
  --display-name="ORAXXXX-Bucket" \
  --source=ora-datastream-XXXX --destination=gcs-oracle-XXXX \
  --oracle-source-config=oracle-source-config.json \
  --gcs-destination-config=gcs-dest-config.json \
  --backfill-all
```

# 4. Pause

```
gcloud datastream streams update oraXXXX-bucket \
  --location=europe-west3 --state=PAUSED --update-mask=state
```

# 5. Resume

```
gcloud datastream streams update oraXXXX-bucket \
  --location=europe-west3 --state=RUNNING --update-mask=state
```



# 6. Delete stream

```
gcloud datastream streams delete oraXXXX-bucket --location=europe-west3
```

# 7. Delete connection profiles

```
gcloud datastream connection-profiles delete ora-datastream-XXXX --location=europe-west
```

```
gcloud datastream connection-profiles delete gcs-oracle-XXXX --location=europe-west3
```

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